X measured, low noise

X + Y measured, low noise

X + Y measured, high noise

A

B

C

Concentration

time

X (True simulation)
X (Measurement)
X (Fit, known)
X (Fit, param)
X (Fit, UDE)
Y (True simulation)
Y (Measurement)
Y (Fit, known)
Y (Fit, param)
Y (Fit, UDE)

D

E

F

Max. likelihood

log(d)

Profile (known)
Profile (param.)
Profile (UDE)
d (true)
95% conf. interval

G

H

I

A(Y;p)

Y

$A(Y;\boldsymbol{p}) = 1.1\frac{Y^3}{2.0^3 + Y^3}$

$A(Y;\boldsymbol{p}) = v\frac{Y^n}{K^n + Y^n}$

$A(Y;\boldsymbol{p}) = U(Y;\boldsymbol{\theta})$